



1 DSA

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1. The arrangement of data so that data can be stored and accessed. The efficiency of the data structure depends upon the conditions of that particular data structure.

Example Array, Link List, Structure, Queue etc.

2. A data structure is a collection of basic data types combined according to the rules of the programming language to create a new user-defined data types.

Storage Structures

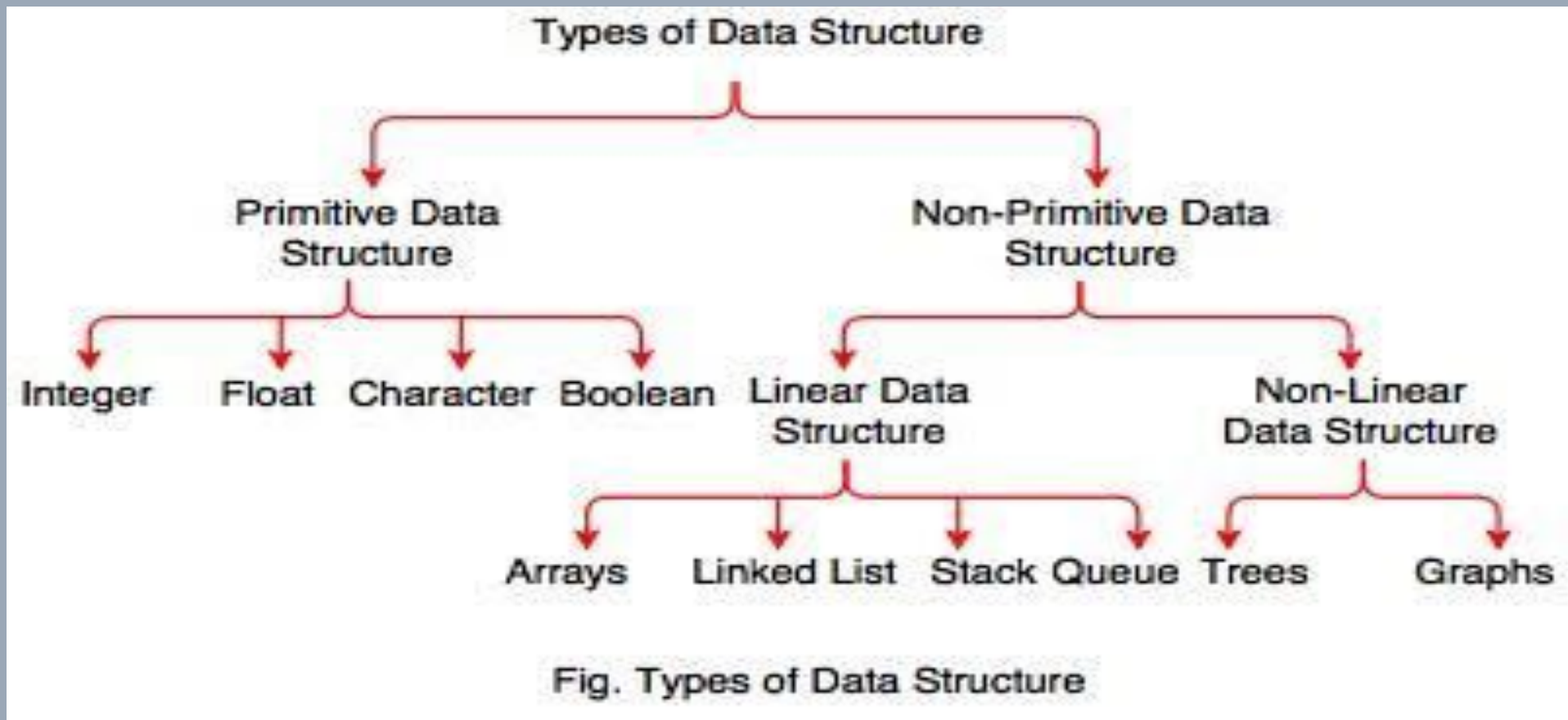
The representation of a particular data structure in the computer memory is called the storage structures.

✓ **Nonlinear data structures**

The data structures whose elements are arranged in non linear or non sequence form are called non linear data structure (Example: trees and graphs).

✓ **linear data structures**

The data structures whose elements are arranged in linear or sequence form are called linear data structure (Example: Array, Link List, Stack Queue etc.)



-DSA

Terminologies

Data:

Raw fact or material known as data.

Data Item:

A data item refers to a single unit of value. Example Student Address. The data item divided into 2 categories.

- i) **Group Item:** Data items that are divided into sub items are called group items
Example Student address can be divided into street#, house # and city.
- ii) **Elementary Item:** The data items that are not further subdivided into subgroups are called elementary item. Example Roll No, employee card no etc.

Terminologies

Entity:

Any object that exist and can be distinguished from other objects. Like hospital, college and any organization.

Entity Set:

A collection of entitles of the same type is called entity set. The collection of all the employees from the employee table at a particular instant of time is an example of an entity set.

Field: A single elementary unit of information representation on the attributes of entity.

A set of one or more related characters is called field , it is the smallest unit of data that can be accessed by user.

Terminologies

Record: A record is a collection of field values of a given entity. A set of related field used as a single unit called record. There are two types of records :

- i) **Variable length Records:** Records in which numbers of attributes are not same. Example data entered about student may vary in their courses or code.
- ii) **Fixed Length Records:** That type of record in which all records contain same number of data items that contain fixed space in memory is called fixed length record.

File: A collection of related records used as a single unit is called file. A file is also known as data set.

Real life vs Data Structures

Real Life

Attributes → Entity → Entity Set

Data Structure

Fields → Records → Files

Operations on Data Structure

A data structure is created to define and process complex data. The data in a data structure is processed using certain operations.

- ✓ Inserting: Adding new data items into a data structure is called inserting.
- ✓ Deleting: Removing new data items into a data structure is called Deleting.
- ✓ Searching: Finding specific data items in a data structure is called Searching.
- ✓ Traversing: Accessing each record or item in a data structure exactly once for processing is called traversing. It is also called visiting.
- ✓ Sorting: Arranging data items in a data structure into a specific order is called sorting.
- ✓ Merging: Combining two lists of data items into a single data list is called merging.

The operation through which data structure is created is called the **creation** operation. Similarly , the operation through which the created data structure is destroyed is called **destruction** operation. The operations on the data structures are represented through algorithms or by directly using the programming language.

Algorithm

The step by step procedure to solve a particular problem is called algorithm. The algorithms notations are used to represent different steps in solving a problem. It is the general solution of the problem. After solving the general problem, the program is written in a programming language by following the given algorithm.

- 1) Name of algorithm
- 2) Steps
- 3) Comments
- 4) Variable Name
- 5) Operators
- 6) Assignment Statement

Classification of Algorithm

1. Recursion or Iteration

A recursive algorithm is one that calls itself repeatedly until a base condition is satisfied. It is a common method used in functional programming languages like C, C++, etc.

2. Procedural or Declarative (non-Procedural)

In declarative programming languages, we say what we want without having to say how to do it.

With procedural programming, we have to specify the exact steps to get the result. For example, SQL is more declarative than procedural, because the queries don't specify the steps to produce the result. Examples of procedural languages include C, PHP, and PERL.

Classification of Algorithm

3. Serial or Parallel or Distributed-

In general, while discussing the algorithms we assume that computers execute one instruction at a time. These are called serial algorithms.

Parallel algorithms take advantage of computer architectures to process several instructions at a time. They divide the problem into subproblems and serve them to several processors or threads. Iterative algorithms are generally parallelized.

If the parallel algorithms are distributed to different machines then we call such algorithms distributed algorithms.

Classification of Algorithm

4. Deterministic or Non-Deterministic-

Deterministic algorithms solve the problem with a predefined process, whereas non-deterministic algorithms guess the best solution at each step through the use of heuristics.

5. Exact or Approximate-

The algorithms for which we are able to find the optimal solutions are called exact algorithms. In computer science, if we do not have the optimal solution, we give approximation algorithms.

Classification of Algorithm

6. Greedy Method-

Greedy algorithms Work in stages. In each stage, a decision is made that is good at that point, without bothering about the future consequences. Generally, this means that some local best is chosen. It assumes that the local best selection also makes for the global optimal solution.

7. Divide and Conquer

Divide: Breaking the problem into subproblems that are themselves smaller instances of the same type of problem.

Recursion: Recursively solving these subproblems.

Conquer: Appropriately combining their answers.

Examples: merge sort and binary search algorithms.

Thank You

I wish for your better Career!